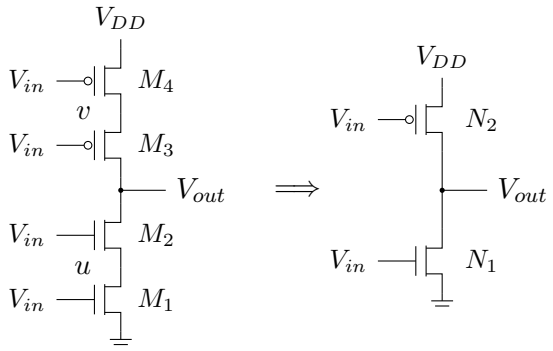
Introduction to VLSI Design, 2020

Homework 1: complete worked solution

The Boolean expressions below were checked against the rendered assignment. In Q3(a), the overbar covers the entire expression; losing that bar changes the required circuit. All body terminals are tied to their normal rails unless the ideal theory explicitly neglects body effect.

1. MOS theory

Let u be the node between the two NMOS devices and v the node between the two PMOS devices. M_1 is nearest ground, M_2 nearest the output; M_4 is nearest V_{DD}, M_3 nearest the output. All gates connect to V_{in}.



(a) M_1 and M_4 cannot be saturated while their corresponding series paths carry nonzero current. For M_1 to saturate, $u \geq V_{in} - V_{tn}$. Then

$$V_{GS,2} = V_{in} - u \leq V_{tn},$$

so M_2 is off (at equality, its ideal overdrive is zero). This contradicts nonzero series current. Thus conducting M_1 must be in the linear region. Likewise, saturation of M_4 requires

$$V_{DD} - v \geq V_{DD} - V_{in} - |V_{tp}| \quad \Longrightarrow \quad v \leq V_{in} + |V_{tp}|.$$

Then $V_{SG,3} = v - V_{in} \leq |V_{tp}|$, so M_3 is off. Conducting M_4 must therefore be linear. M_2 and M_3, the devices adjacent to the output, can saturate. A cutoff device is not called a conducting saturated device here.

(b) Equivalent device sizes. Two equal, common-gate series devices add their channel lengths in the ideal long-channel, body-effect-free model:

$$\boxed{(W/L)_{N_1} = (W/L)_{N_2} = \tfrac{1}{2}(W/L).}$$

For example, keep W and use $2L$, or keep L and use $W/2$. In the linear region the two voltage drops sum and yield half the single-device conductance; the same effective $\beta/2$ follows by eliminating the internal node in the square-law saturation equations. This is the ideal static characteristic equivalence intended by the problem. Internal diffusion capacitances and short-channel/body effects prevent a claim of identical physical transient behavior.

2. Multiplexer-based design

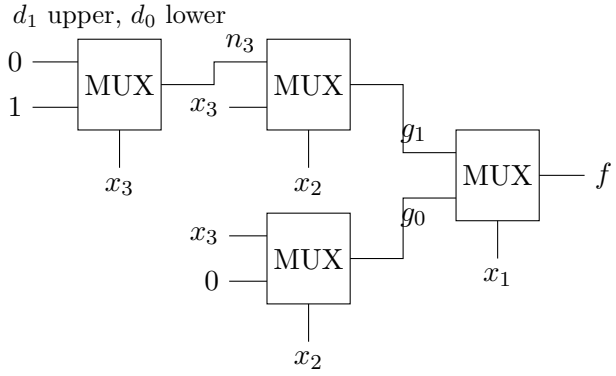
Use $\mathrm{MUX}(s, d_0, d_1) = \bar{s}d_0 + sd_1$. Shannon expansion on x_1 gives

$$f = \bar{x}_1(x_2x_3) + x_1(\bar{x}_2x_3 + x_2\bar{x}_3).$$

The following four 2-to-1 multiplexers implement the function using only true inputs and constants:

$$\begin{aligned} n_3 &= \mathrm{MUX}(x_3, 1, 0) = \bar{x}_3, \\ g_0 &= \mathrm{MUX}(x_2, 0, x_3) = x_2x_3, \\ g_1 &= \mathrm{MUX}(x_2, x_3, n_3) = x_2 \oplus x_3, \\ f &= \mathrm{MUX}(x_1, g_0, g_1). \end{aligned}$$

If $\bar{x}_3$ is already available, only the final three MUXes are required.



x_1	x_2	x_3	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

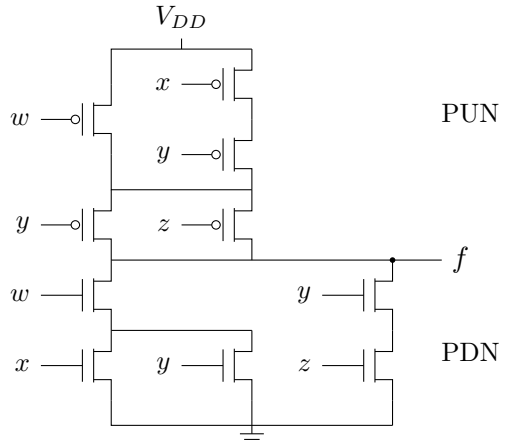
The output is 1 exactly when two of the three inputs are 1.

3(a). Static CMOS: $f = \overline{w(x+y)+yz}$

The pull-down network (PDN) implements $G = w(x+y)+yz$. Its two parallel paths are n_w in series with $(n_x \parallel n_y)$, and n_y in series with n_z. The complementary pull-up network (PUN) is

$$\bar{G} = [\bar{w} + \bar{x}\bar{y}][\bar{y} + \bar{z}].$$

Thus use $p_w \parallel (p_x \text{ series } p_y)$, in series with $p_y \parallel p_z$. All gate labels below are true inputs; PMOS devices turn on when their labeled input is 0. Repeated y labels denote distinct transistors. There are 5 PMOS and 5 NMOS devices.



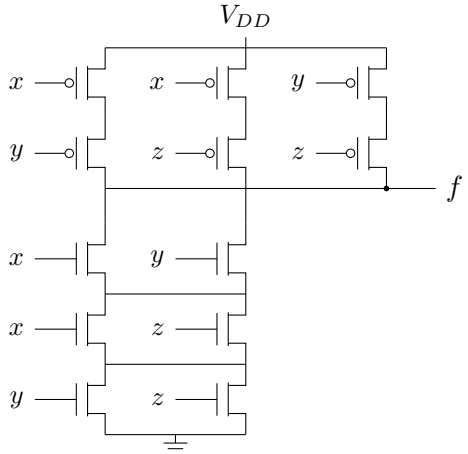
Series connections implement AND in a conducting network and parallel connections implement OR. De Morgan's law makes the networks mutually exclusive for every binary input combination: if $G = 1$, PDN conducts and PUN does not; if $G = 0$, PUN conducts and PDN does not. No input inverter is needed.

3(b). Static CMOS: $f = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z}$

Only true literals are supplied, so form the complement of the desired function for the NMOS gates:

$$\bar{f} = (x + y)(x + z)(y + z) = xy + xz + yz.$$

Use three series stages, each containing two parallel NMOS devices, for the PDN. The dual PUN has three parallel branches, each containing two series PMOS devices. It contains 6 PMOS and 6 NMOS devices and needs no external complemented inputs.



This function is 1 for input Hamming weight 0 or 1 and 0 for weight 2 or 3. The product-of-sums network follows the supplied handwritten solution. An alternative PDN is three parallel two-NMOS stacks (xy, xz, yz), with its dual PUN; both implement the same function and use 12 devices.

Verification and source reconciliation

The supplied `HW1_answer.pdf` is a two-page handwritten submission by J.-M. Wu (student ID 107501503). Its answers agree on M_1/M_4, the half-size equivalence, the Shannon construction, and the CMOS networks. The present report expands the saturation proof, distinguishes ideal static equivalence from real capacitance behavior, and supplies an all-MUX implementation of $\bar{x}_3$.

Run `python verify_logic.py`. It checks all 8 MUX cases, all 16 cases of Q3(a), and all 8 cases of Q3(b), including complementary PUN/PDN conduction. The checked truth table and summary are `truth_tables.csv` and `verification.json`. This logic verification is not an electrical transistor simulation.